Zengtao Liang

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EDUCATION

University of Florida, Gainesville, Florida — Expected Graduation: May 2027

BS in Computer Science — *GPA:3.97/4.00*

Relevant Coursework: Operating Systems, Computer Structure and Algorithms, Intro to Software Engineering, Info and Database Systems, Natural Language Processing

Awards: Dean's List

West Boca Community High School, Boca Raton, Florida — Graduation Year:2023

*Awards:*Honors Scholar Award, Pathfinder nominee for Math

EXPERIENCE

Math Tutor, Boca Raton, FL — June 2020 – June 2023

- Provided one-on-one tutoring in mathematics from elementary to high school level, including Algebra, Geometry, and Calculus
- Tracked student progress, identified areas for improvement, and adjusted lesson plans accordingly

Chinese Learning Assistant, Boca Raton, FL — June 2021 – June 2023

- Assisted children of various ages in learning Mandarin Chinese, focusing on speaking, listening, reading, and writing skills
- Maintained regular communication with parents to provide progress updates and suggest home practice strategies

PROJECTS

FUSE WAD File System, C++ — August 2025

- Designed and implemented a user-space file system using FUSE to mount and navigate WAD archive files as a hierarchical directory structure
- Built a custom WAD parser to read and interpret on-disk descriptors, directory entries, and lump metadata
- Diagnosed and resolved low-level I/O and descriptor offset bugs that caused corrupted reads and improper metadata mapping
- Implemented directory traversal, file extraction, and writable operations with correct inode metadata persistence

Task Management Web App, MERN Stack — May 2025

- Built a full-stack task management application using **MongoDB, Express.js, React.js, and Node.js (MERN)**
- Designed and implemented RESTful APIs for task CRUD operations and deadline tracking
- Developed a responsive React frontend with dynamic state management and real-time UI updates
- Integrated MongoDB for persistent data storage with structured schema design

Lua Interpreter, Rust — May 2024

- Implemented core components of a Lua interpreter in Rust, including lexical analysis and tokenization
- Designed foundational parsing infrastructure to support future AST construction and execution

SKILLS AND INTERESTS

- *Computer Skills:* Java, JavaScript, Python, C++, Rust, SQL, R, HTML, React.js, Node.js, MongoDB, Unity, Blender
- *Language Skills:* English – fluent, Chinese - fluent
- *Certifications:*
  - *Microsoft Technology Associate: Introduction to Java*
  - *Adobe Certified Professional: Adobe Photoshop 2018*
  - *Microsoft Office Specialist: Excel, Word, and PowerPoint*